

A MACHINE-CHECKED RE-ENTRANCY OBSTRUCTION FOR SUPERVISOR–WORKER ORCHESTRATION THE COMPOSITION CLASS $[\omega] = \varepsilon$ IN LEAN

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ABSTRACT. An orchestration of AI agents is a directed container, and welding two agents that share a resource into one interleaved system is a Zappa–Szép product of their handoff categories — the distributive-law construction for directed containers being Ahman–Uustalu’s. Such a product need not exist, and its non-existence is a degree-two cohomology class $[\omega] \in H^2(\text{Sk}_C; \mathcal{D})$ on the handoff category. We take the smallest faithful instance — a supervisor dispatching to a worker, parametrized by a one-bit turn token whose mutation bit ε marks whether the worker writes to the supervisor’s shared state during a pending call — and package its obstruction as a finite $\mathbb{F}_2$ linear-algebra computation. The output is a short, self-contained Lean 4 development, sorry-free and free of external libraries, whose theorems certify that the obstruction class equals the token-mutation bit on the nose, $[\omega] = \varepsilon$, and hence that the joint agent exists if and only if $\varepsilon = 0$. Unprotected re-entrancy — the DAO-class failure mode — is thus not a heuristic smell but a machine-checked nonzero generator of $H^2 \cong \mathbb{Z}/2$. We state precisely what the formalisation certifies and what remains a pen-and-paper input.

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1. INTRODUCTION

A concrete failure. In 2016 an attacker drained roughly a third of the Ethereum DAO’s funds through a single structural flaw: a contract called out to an external party *before* updating its own bookkeeping, and the callee re-entered the still-open call and spent against stale state. *Unprotected re-entrancy* — a callee mutating the caller’s shared state during a pending call — has since become the canonical smart-contract hazard, and the same shape recurs whenever a supervisor dispatches work to a sub-agent that may call back before returning. The question this note answers is whether that hazard has an *invariant*: a mathematical object that is zero exactly when the composition is safe and nonzero exactly when it is not, and which can be *computed and checked* rather than argued informally.

Agents are containers, composition is a Zappa–Szép product. The setting is the polynomial-functor (container) model of interaction, due to Spivak [1] and developed for agent orchestration by Ghani at Kodamai. An agent that may issue queries from a set S and, for each query s , receive a reply from $P(s)$ is the container $p = (S, P)$, i.e. the functor $\sum_{s \in S} y^{P(s)}$. A *handoff topology* — who may delegate to whom, staying put, concatenating handoffs — is a small category, equivalently a directed container in the sense of Ahman–Uustalu [3]. Running two orchestrations that share a resource is not composition *along* one fixed topology (which a functor already models and which therefore cannot fail); it is the *interleaving* of two, and the interleaving of two directed containers is a *Zappa–Szép product* $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$. That a distributive law of directed containers assembles such a joint container — generalising the Zappa–Szép product of monoids, its matched pair recording the two mutual actions — is a theorem of Ahman–Uustalu [4]; we take the construction from them and claim none of it.

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The gap, and the invariant. A Zappa–Szép product *can fail to exist*, and Ahman–Uustalu’s account — which is our source for the construction — builds no measure of that failure: composition simply exists when a distributive law does. Supplying the missing invariant is the author’s contribution elsewhere: the existence of the complementary factor is governed by a single degree-two cohomology class,

$$\mathcal{C} \bowtie \mathcal{D} \text{ exists} \iff [\omega] = 0 \in H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D}),$$

cohomology of the handoff category’s skeleton with coefficients in the reply (direction) functor $\mathcal{D}$ [6, 5]. The cohomology itself is classical [9, 10, 11]; the delta over Ahman–Uustalu is the *identification* of the orchestration composite with $\mathcal{C} \bowtie \mathcal{D}$ and hence of its obstruction with this class.

The result, and what is new here. The present note is not the proof that $[\omega]$ is the right invariant — that is the author’s analytic argument [7], whose spine we recall only as far as needed. What is new here is a *machine-checked* certificate for the smallest faithful instance. Fix a supervisor S dispatching to a worker W that returns to R , carrying a one-bit turn token. A single design bit $\varepsilon \in \mathbb{Z}/2$ records whether the worker’s re-entrant outcome *mutates* the token ($\varepsilon = 1$, unprotected re-entry) or *fixes* it ($\varepsilon = 0$, a lock). The analytic argument reduces the categorical obstruction to a tiny cochain complex over the two-element field $\mathbb{F}_2$; the class living in that complex is what we certify. Our main statement is the following, proved in Lean 4 with no `sorry` and no external library (Section 4, Theorem 4.1).

Theorem 1.1 (machine-checked). *Let $\varphi(a, b) = b - a$ be the class map on the 2-cochains $C^2 \cong \mathbb{F}_2^2$, whose kernel is the coboundary subgroup B^2 , so that φ realises $H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D}) = C^2/B^2 \cong \mathbb{F}_2$. Let $\omega_\varepsilon = (0, \varepsilon)$ be the defect cocycle of the natural transversal. Then*

$$\varphi(\omega_\varepsilon) = \varepsilon, \quad \omega_\varepsilon \in B^2 \iff \varepsilon = 0.$$

Equivalently: the composition class $[\omega]$ equals the token-mutation bit, and the joint agent $\mathcal{C} \bowtie \mathcal{D}$ exists if and only if $\varepsilon = 0$.

The reader should note exactly what changes relative to the author’s broad orchestration paper [8]. There the identity $[\omega] = \varepsilon$ is established analytically and the formalisation is a corollary cross-check; *here the Lean development is the deliverable*, and the note is built to state, quote, and delimit it. The verification boundary is drawn honestly in Section 5: the formalisation certifies the finite $\mathbb{F}_2$ class computation — the step that turns “a cochain complex” into “ $[\omega] = \varepsilon$ ” — and takes the reduction of the categorical obstruction *to* that complex as a cited, pen-and-paper input.

Why the degree matters. One structural remark orients the reader against the neighbouring literature. Multi-agent-systems work places cellular-sheaf obstructions of degree H^0 (consensus, Nash equilibria) and H^1 (identifiability) on the agents’ *communication* graph, asking whether agents currently agree [13, 14]. The container obstruction sits one degree up, in H^2 , on the *handoff* category, and asks the prior question: can the agents be fused into one consistent joint agent at all? Re-entrancy is the smallest place where the answer is no, and $[\omega] = \varepsilon$ is why.

Outline. Section 2 fixes the container / directed-container / Zappa–Szép vocabulary and the parametrized supervisor–worker category $\mathcal{K}_\varepsilon$. Section 3 records the finite $\mathbb{F}_2$ complex that the analytic reduction produces, and the class $\omega_\varepsilon = (0, \varepsilon)$. Section 4 presents the Lean development — the load-bearing artifact — quoting its definitions and theorems verbatim and stating precisely what they certify. Section 5 draws the verification boundary and disposes honestly of a demoted side-reading. Section 6 closes.

2. PRELIMINARIES

We include only what the statement and its Lean certificate use. Containers, the directed-container / small-category identification, and the Zappa–Szép construction are all prior art and are cited as such; the parametrized category $\mathcal{K}_\varepsilon$ is the minimal faithful model of the supervisor–worker pattern.

Containers and directed containers. A *container* $p = (S, P)$ is a set S of shapes together with, for each $s \in S$, a set $P(s)$ of positions; it represents the polynomial functor $\sum_{s \in S} y^{P(s)}$ [1, 2]. A *directed container* equips (S, P) with a root, a sub-position operation, and a position-concatenation, and Ahman–Uustalu’s theorem is that directed containers are exactly small categories: objects are shapes, morphisms out of s are positions $P(s)$, identities are roots, and composition is concatenation [3]. We use the category picture throughout and call it the *handoff category*.

Zappa–Szép products. Let $\mathcal{K}$ be a small category and $\mathcal{D} \subseteq \mathcal{K}$ a wide subcategory (same objects, some morphisms). A *Zappa–Szép product* $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ is a second wide subcategory $\mathcal{C}$ such that every morphism of $\mathcal{K}$ factors *uniquely* as a $\mathcal{C}$ -morphism followed by a $\mathcal{D}$ -morphism — a strict factorization system, equivalently a distributive law $\lambda : \mathcal{D}\mathcal{C} \Rightarrow \mathcal{C}\mathcal{D}$ satisfying the matched-pair axioms. For directed containers this is the composition construction of Ahman–Uustalu [4]: it generalises the Zappa–Szép product of two monoids, and its data is a matched pair of mutual actions. The point for us is that, given $\mathcal{D}$, the complementary factor $\mathcal{C}$ *may or may not exist*; that existence is what the obstruction measures.

The parametrized supervisor–worker category. Fix three roles: S (supervisor), W (a worker running), R (a return point). The supervisor carries a one-bit *turn token*; toggling it is an involution τ with $\tau^2 = \text{id}$, so $\text{End}(S) = \{\text{id}_S, \tau\} \cong \mathbb{Z}/2$. The generating handoffs are

$$p, p\tau : S \rightarrow W \quad (\text{dispatch in turn-state } 0 \text{ resp. } 1), \quad s, s_2 : W \rightarrow R \quad (\text{two worker outcomes}),$$

with composites $q, q\tau : S \rightarrow R$ and right $\mathbb{Z}/2$ -action $p \cdot \tau = p\tau$, $q \cdot \tau = q\tau$. The prescribed right factor is the supervisor-internal (token) part, $\mathcal{D} = \{\text{id}_S, \tau, \text{id}_W, \text{id}_R\}$, a wide subcategory whose only nontrivial vertex group is $G_S \cong \mathbb{Z}/2$; the left factor $\mathcal{C}$ — the genuine dispatch/handoff part — is what a Zappa–Szép product would have to supply.

Definition 2.1 (the family $\mathcal{K}_\varepsilon$). For $\varepsilon \in \mathbb{Z}/2$ let $\mathcal{K}_\varepsilon$ be the category above with the single length-two composite fixed by

$$s \circ p = q, \quad s_2 \circ p = q \cdot \tau^\varepsilon,$$

where s is “the worker returns normally” and s_2 is “the worker re-enters” — calls back into the supervisor before returning. Thus $\varepsilon = [s_2 \circ p = q\tau]$ records whether the re-entrant outcome *mutates* the token ($\varepsilon = 1$, unprotected re-entry) or *fixes* it ($\varepsilon = 0$, state-protected re-entry / a lock).

Each $\mathcal{K}_\varepsilon$ is a genuine category with $\mathcal{D}$ wide [7]; the reading is exact, in that re-entrancy — the callee mutating shared caller state during a pending call — is precisely the s_2 -branch dragging the token when $\varepsilon = 1$.

3. THE OBSTRUCTION AS A FINITE $\mathbb{F}_2$ COMPLEX

The categorical obstruction of [6] is, for $\mathcal{K}_\varepsilon$, computed by an explicit orbit/cochain calculation in [7]. We do not reproduce that reduction; we record its output, because that output is exactly what the Lean development certifies. The single fact we import is:

Proposition 3.1 (analytic reduction; [7]). *For each $\varepsilon \in \mathbb{Z}/2$ the family $\mathcal{K}_\varepsilon$ satisfies the freeness condition (L) and the vertex-group hypothesis of the obstruction theorem, its skeleton $\text{Sk}_{\mathcal{K}}$ is the branch $S \rightarrow W \rightrightarrows R$ (independent of ε), and the normalized cochain complex of $H^\bullet(\text{Sk}_{\mathcal{K}}; \mathcal{D})$ is*

$$C^1 \cong \mathbb{F}_2^2, \quad C^2 \cong \mathbb{F}_2^2, \quad C^3 = 0,$$

with coordinates $h([p]), h([q])$ on C^1 and $\omega([s], [p]), \omega([s_2], [p])$ on C^2 ; the coboundary is $\delta^1 h = (h[p] - h[q], h[p] - h[q])$; and the defect 2-cocycle of the natural transversal $T = \{p, q, s, s_2\}$ is $\omega_T(\varepsilon) = (0, \varepsilon)$.

Everything after Proposition 3.1 is finite linear algebra over $\mathbb{F}_2$, and it is this part that we formalise. Write $C^2 = \mathbb{F}_2^2$. Since $C^3 = 0$ we have $Z^2 = C^2$. The coboundary image is the diagonal,

$$B^2 = \text{im } \delta^1 = \{(t, t) : t \in \mathbb{F}_2\} = \langle (1, 1) \rangle,$$

so $H^2 = C^2/B^2 \cong \mathbb{F}_2$, and the quotient is realised by the class map

$$\varphi : \mathbb{F}_2^2 \rightarrow \mathbb{F}_2, \quad \varphi(a, b) = b - a = a \oplus b,$$

whose kernel is exactly B^2 . The defect cocycle is $\omega_\varepsilon := \omega_T(\varepsilon) = (0, \varepsilon)$, so $\varphi(\omega_\varepsilon) = \varepsilon$: its class is trivial for $\varepsilon = 0$ and the nonzero generator for $\varepsilon = 1$. Chaining this with the obstruction theorem gives the dichotomy

$$\mathcal{C} \bowtie \mathcal{D} \text{ exists} \iff [\omega_\varepsilon] = 0 \iff \varepsilon = 0.$$

This is Theorem 1.1; the next section certifies its finite core.

Remark 3.2 (reading the class). The two coordinates of ω are the two worker outcomes; the coboundary diagonal is the freedom to relabel which token-state is “canonical.” The class is nonzero precisely because the two outcomes disagree on that relabelling by one token-flip — the re-entrant branch’s side effect. This is the categorical form of “a flat bundle is trivial iff its holonomy vanishes”: each partial handoff completes freely, but the completions fail to glue into a global serialized agent by a single $\mathbb{Z}/2$ monodromy around the dispatch/return loop.

4. THE LEAN FORMALISATION

This section is the note’s load-bearing artifact. The finite $\mathbb{F}_2$ computation of Section 3 is transcribed into Lean 4 and machine-checked. The development is sorry-free and uses no external library — in particular no Mathlib — matching the pure-core style of the author’s container development (`Basic.lean`, `ZappaSzep.lean`); this is possible because the entire computation is finite, so the substantive facts close by `decide` or by case exhaustion. The file is `Containers/Reentrancy.lean`. The listings below transcribe it faithfully, with Unicode replaced by ASCII for typesetting: `exists` is $\exists$, `<->` is $\leftrightarrow$, `<-` is the reverse rewrite $\leftarrow$, `x` is the product $\times$, `e` is ε , and `<·>` is the anonymous constructor.

The model. The two-element field $\mathbb{F}_2$ is modelled as `Bool` with `false` = 0, `true` = 1, and `xor` as addition (which, in characteristic two, is also subtraction). The objects of Section 3 become the following declarations.

```
abbrev Z2 := Bool                -- F2, with xor = + = -
abbrev C2 := Z2 x Z2             -- the 2-cochains C2, coords omega([s],[p]), omega([s2],[p])

def d1 (t : Z2) : C2 := (t, t)   -- delta^1 reparametrised: image is the diagonal
def InB2 (x : C2) : Prop := exists t : Z2, d1 t = x  -- membership in B2 = im delta^1
def omega (e : Z2) : C2 := (false, e)  -- transversal defect omega_T = (0, e)
def phi (x : C2) : Z2 := xor x.1 x.2  -- class map (a,b) /-> b - a
```

Here `Z2`, `C2`, `d1`, `InB2`, `omega`, `phi` are $\mathbb{F}_2$, C^2 , δ^1 , B^2 , ω_T , φ respectively.

The certified theorems. Three facts establish $H^2 \cong \mathbb{F}_2$ via φ : that φ kills the coboundary, that its kernel is *exactly* the coboundary (so it descends to an injection $H^2 \hookrightarrow \mathbb{F}_2$), and that it is surjective. In Lean:

```

theorem phi_d1 (t : Z2) : phi (d1 t) = false          -- phi kills B2
theorem phi_ker_eq_inB2 (x : C2) : phi x = false <-> InB2 x  -- ker phi = B2 (H2
  injects into F2)
theorem phi_surjective (y : Z2) : exists x : C2, phi x = y    -- phi onto (H2 = F2)

```

The main identity and the dichotomy are then two lines:

```

-- Class form of [omega] = e: the class of the defect cocycle (0,e) is exactly e.
theorem phi_omega (e : Z2) : phi (omega e) = e := by
  cases e <;> rfl

-- Membership form of [omega] = 0 <-> e = 0: the defect is a coboundary iff e = 0,
-- iff the Zappa-Szep product K_e = C /></ D exists.
theorem omega_inB2_iff_zero (e : Z2) : InB2 (omega e) <-> e = false := by
  rw [<- phi_ker_eq_inB2, phi_omega]

```

and the two endpoints are recorded as sanity checks that close by `rfl`:

```

example : phi (omega true) = true := rfl          -- e = 1: nonzero class, product does NOT
  exist
example : InB2 (omega false) := <false, rfl>      -- e = 0: trivial class, product exists

```

Theorem 4.1 (the certified statement). *In Containers/Reentrancy.lean, sorry-free and without external libraries, the following are machine-checked:*

- *phi_ker_eq_inB2* and *phi_surjective*, together giving the isomorphism $H^2 = C^2/B^2 \cong \mathbb{F}_2$ realised by φ ;
- *phi_omega* : $\varphi(\omega_\varepsilon) = \varepsilon$ — the class form of $[\omega] = \varepsilon$;
- *omega_inB2_iff_zero* : $\omega_\varepsilon \in B^2 \iff \varepsilon = 0$ — the membership form of $[\omega] = 0 \iff \varepsilon = 0$, i.e. the joint agent $C \bowtie D$ exists iff the worker fixes the token.

Consequently, on the finite complex of Proposition 3.1, the obstruction class equals the token-mutation bit, and unprotected re-entrancy ($\varepsilon = 1$) is the nonzero generator of $H^2 \cong \mathbb{Z}/2$.

The proofs are as short as they look. `phi_omega` is a two-case split closing by `rfl`; `omega_inB2_iff_zero` rewrites by $\ker \varphi = B^2$ and then by $\varphi(\omega_\varepsilon) = \varepsilon$; `phi_ker_eq_inB2` is a four-case exhaustion over $(a, b) \in \mathbb{F}_2^2$. The brevity is the point: once the categorical obstruction is reduced to this complex, the identity $[\omega] = \varepsilon$ is a finite fact that a proof assistant settles with no room for hand-waving.

5. DISCUSSION

The verification boundary, drawn honestly. It matters to say exactly what is and is not certified. The Lean development checks the *finite $\mathbb{F}_2$ class computation*: given the complex of Proposition 3.1 — $C^2 \cong \mathbb{F}_2^2$, the coboundary equal to the diagonal, and the defect cocycle equal to $(0, \varepsilon)$ — it proves $H^2 \cong \mathbb{F}_2$, $\varphi(\omega_\varepsilon) = \varepsilon$, and the membership dichotomy. It does *not* certify the *reduction* of the categorical obstruction to that complex: that $\mathcal{K}_\varepsilon$ satisfies (L) and the vertex-group hypothesis, that $\text{Sk}_\mathcal{C}$ is the stated branch, and that the transversal defect is $(0, \varepsilon)$ are Proposition 3.1, established by hand in [7] and imported here as input. So the honest reading of Theorem 4.1 is: *the last mile — from cochain complex to $[\omega] = \varepsilon$ — is machine-checked; the first mile — from category to cochain complex — is a cited pen-and-paper theorem*. Formalising the first mile (the orbit category, the freeness argument, the defect computation) is the natural next Lean target and is not attempted here.

A demoted side-reading. An earlier framing of this material distinguished a “total” from a “partial” composition and predicted that the two would be separated by a degree-*one* invariant. That prediction is withdrawn. The correct statement is the classical distinction between an *absolute* and a *classified* obstruction — Gerstenhaber’s “absolute versus classified” dichotomy for deformation/obstruction problems [12] — and it carries no novelty and no separate H^1 degree here; the single degree-two class $[\omega]$ of Section 3 is the whole invariant. We flag the point only to retire the earlier prediction cleanly.

Relation to the broad orchestration paper. This note is deliberately narrow. The wider case for reading agent orchestration as directed-container composition — the honestly-graded dictionary, the four-regime table (independent product, the non-abelian S_3 interleave, and both re-entry rows), and the degree-axis contrast with sheaf-Laplacian multi-agent methods — is the author’s broad paper [8], where $[\omega] = \varepsilon$ is proved analytically and the present formalisation appears as a cross-check. The contribution of *this* note is the inversion of emphasis: the machine-checked certificate is foregrounded and delimited, so that a formal-methods reader can point to a sorry-free file and a category theorist can see precisely which finite fact it settles.

6. CONCLUSION

Unprotected re-entrancy — a worker writing to the supervisor’s shared state during a pending call, the DAO-class failure — is a nonzero degree-two cohomology class on the handoff category, and on the minimal faithful model that class equals the token-mutation bit on the nose. The identity $[\omega] = \varepsilon$, and with it the dichotomy that the joint agent exists if and only if the worker fixes the token, is now a sorry-free Lean 4 theorem depending on no external library. What the formalisation buys is not a new mathematical fact but a *located* one: the composability of two shared-resource agents is decided by a finite object a proof assistant can check, and the check is the last mile of an argument whose first mile — the reduction from category to complex — is the clearly marked pen-and-paper input and the obvious next formalisation target. Beyond that lie larger handoff topologies, where the obstruction is a class in a genuinely larger H^2 , and the ∞ -analogue, where a re-entrant tool-call is itself an orchestration.

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